7(a)	The *magnetic flux density at a point* is defined as the force acting per unit current per unit length of a conductor when the conductor is placed at right angles to the field.	B1
(b)	$B_R = B_A - B_B = \frac{\mu_o I_A}{2\pi d} - \frac{\mu_o I_B}{2\pi d}$ $= \frac{4\pi \times 10^{-7}}{2\pi}\left(\frac{0.32}{0.06} - \frac{0.080}{0.06}\right)$ $= 8.0 \times 10^{-7}$ T	C2 A1
(c)	(By Fleming's Left Hand Rule), electromagnetic force on wire is vertically upwards Force on wire = BIL = 0.065 (2.2)0.040 =5.72 x 10^{-3} N By Newton's Third Law, there is an equal and opposite force acting downwards on the magnet. Reading on balance = 1.1772 + 0.00572 = 1.1829 N *[Balance reading must be given to precision of the instrument of 4 dp.]*	B1 B1 B1 A1